

Early Warning Signals in Recurrent Novae: Observations, Toy Models, and Slowly Driven Accretion Systems

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1 Introduction

Recurrent novae are interacting binary systems consisting of a white dwarf accreting material from an evolved companion star. As hydrogen-rich material accumulates on the white dwarf surface, the temperature and pressure at the base of the accreted envelope gradually increase until thermonuclear ignition occurs. This ignition triggers a thermonuclear runaway (TNR), producing a nova eruption and a sudden increase in luminosity.

Although the physical mechanism responsible for nova eruptions is relatively well understood, predicting the timing of these events remains a major challenge. Most observational studies focus on the eruption itself, whereas comparatively little attention has been devoted to identifying signatures that may appear before the transition occurs.

In recent years, the concept of Early Warning Signals (EWS) has emerged as a powerful framework for detecting approaching critical transitions in complex dynamical systems [Scheffer et al., 2009]. EWS are statistical signatures that arise when a system gradually loses stability and approaches a tipping point. Such signals have been observed in ecological systems, climate dynamics, financial markets, and physiological systems.

The first astrophysical application of this framework was presented by George et al. [2020], who reported the presence of EWS prior to the Great Dimming of Betelgeuse. Their results demonstrated that stellar light curves can contain measurable indicators of an approaching transition.

Motivated by this success, we investigate whether similar signatures can be identified in recurrent novae. In particular, we focus on two well-studied systems:

- T Coronae Borealis (T CrB)
- RS Ophiuchi (RS Oph)

These systems provide an interesting comparison because both undergo recurrent nova eruptions but differ significantly in their accretion environments and observational behavior.

The primary objective of this work is to determine whether EWS can be detected before nova eruptions and to identify the physical mechanism responsible for their appearance.

2 Critical Slowing Down and Early Warning Signals

The theoretical basis of Early Warning Signals is the phenomenon known as Critical Slowing Down (CSD).

Consider a dynamical system governed by

$$\frac{dx}{dt} = f(x, \mu), \tag{1}$$

where μ is a control parameter.

Suppose the system possesses a stable equilibrium state x^* satisfying

$$f(x^*, \mu) = 0. \quad (2)$$

A small perturbation around equilibrium may be written as

$$x = x^* + \epsilon. \quad (3)$$

Linearizing around the equilibrium gives

$$\frac{d\epsilon}{dt} = \lambda\epsilon, \quad (4)$$

where

$$\lambda = \left. \frac{df}{dx} \right|_{x=x^*}. \quad (5)$$

The solution is

$$\epsilon(t) = \epsilon_0 e^{\lambda t}. \quad (6)$$

For a stable equilibrium, $\lambda < 0$ and perturbations decay exponentially.

As the system approaches a critical transition, the restoring force weakens and λ approaches zero. Consequently, perturbations require increasingly longer times to decay.

This slowing of recovery is known as Critical Slowing Down.

When stochastic fluctuations are present, CSD produces several observable statistical signatures:

1. Increasing variance
2. Increasing lag-1 autocorrelation
3. Increased long-range temporal correlations
4. Structural changes in recurrence plots

These signatures constitute the Early Warning Signals.

2.1 Variance

For a stochastic autoregressive process

$$y_t = \alpha y_{t-1} + \sigma \eta_t, \quad (7)$$

the variance is

$$\text{Var}(y) = \frac{\sigma^2}{1 - \alpha^2}. \quad (8)$$

As the system approaches a transition, $\alpha \rightarrow 1$, causing the variance to increase.

2.2 Lag-1 Autocorrelation

The lag-1 autocorrelation coefficient is given by

$$ACF(1) = \alpha. \quad (9)$$

Near a critical point, the memory of previous perturbations increases, causing autocorrelation to rise.

2.3 Detrended Fluctuation Analysis

Detrended Fluctuation Analysis (DFA) measures long-range correlations in a time series.

The fluctuation function scales as

$$F(s) \propto s^\beta, \quad (10)$$

where β is the DFA exponent.

An increasing DFA exponent indicates stronger temporal persistence and is therefore considered an EWS indicator.

2.4 Recurrence Quantification Analysis

Recurrence Quantification Analysis (RQA) characterizes the geometry of trajectories in phase space.

The recurrence matrix is defined as

$$R_{ij} = \Theta(\varepsilon - \|x_i - x_j\|), \quad (11)$$

where Θ is the Heaviside function and ε is the recurrence threshold.

Two RQA measures are particularly useful:

- Determinism (DET)
- Laminarity (LAM)

Increasing DET indicates more coherent deterministic structure, while increasing LAM reflects the presence of slowly evolving states. Both quantities are expected to increase near Critical Slowing Down.

3 Observational Analysis

The observational component of this study utilized light-curve data obtained from the American Association of Variable Star Observers (AAVSO).

Because irregular sampling and observational gaps can strongly influence statistical analyses, the data were binned into 10-day intervals. This procedure significantly reduced the impact of data gaps while preserving the long-term variability of the systems.

Rolling-window analysis was then performed on the detrended light curves. Within each window, the following EWS indicators were calculated:

- Variance
- Lag-1 Autocorrelation
- DFA exponent
- Determinism (DET)
- Laminarity (LAM)

The significance of monotonic trends was assessed using Kendall's τ statistic and the modified Mann-Kendall test.

3.1 Results for T CrB

The analysis of T CrB revealed a remarkably consistent pattern.

Multiple indicators displayed statistically significant increasing trends prior to eruption, including:

- Variance
- Lag-1 autocorrelation
- DFA
- DET
- LAM

The simultaneous increase of several independent indicators strongly suggests the presence of Critical Slowing Down.

The results therefore provide evidence that the system undergoes a gradual loss of stability before eruption.

3.2 Results for RS Oph

The same methodology was applied to multiple RS Oph eruption cycles.

Unlike T CrB, RS Oph did not exhibit a similarly robust set of Early Warning Signals.

Some individual indicators occasionally displayed increasing trends; however, these trends were neither consistent across different eruptions nor simultaneously present in all EWS metrics.

The absence of a clear EWS pattern in RS Oph immediately raised an important question:

Why does T CrB exhibit strong Early Warning Signals while RS Oph does not?

Answering this question became the central motivation for the remainder of this study.

4 Initial Hypothesis: Accretion Disc Stability

The first explanation considered was based on differences in the accretion environments of the two systems.

T CrB is believed to possess a large and persistent accretion disc fed through Roche-lobe overflow from its red giant companion.

In contrast, RS Oph primarily accretes material from the stellar wind of its companion star.

Because Roche-lobe overflow produces a relatively stable and continuous supply of material, it appeared plausible that the accretion disc in T CrB could evolve gradually toward a critical state. Such slow evolution would naturally allow Critical Slowing Down to develop.

RS Oph, on the other hand, experiences a more irregular accretion flow. If transitions occur rapidly, the system may not spend sufficient time near marginal stability for EWS to emerge.

This led to our initial hypothesis:

The presence of Early Warning Signals in T CrB is a consequence of its stable accretion disc, whereas the absence of EWS in RS Oph results from a more unstable accretion environment.

5 Failure of the Initial Hypothesis

Although attractive, the disc-stability hypothesis encountered several difficulties.

Theoretical studies by Alexander et al. [2011] and Bollimpalli et al. [2018] indicate that both T CrB and RS Oph are capable of experiencing thermal-viscous disc instabilities.

Consequently, the distinction between a ‘stable’ disc and an ‘unstable’ disc is not sufficient to explain the observational differences.

More importantly, the hypothesis failed to identify a specific dynamical mechanism capable of generating the observed EWS.

The existence of an instability alone does not necessarily imply the presence of Critical Slowing Down. A system may transition abruptly without exhibiting any measurable warning signals.

These considerations forced us to reconsider the problem from a more fundamental perspective.

Rather than asking whether the disc is stable or unstable, we shifted our attention to a different question:

What property of the system determines whether Critical Slowing Down can develop?

To answer this question, we constructed a sequence of simplified toy models designed to isolate the physical ingredients required for the emergence of Early Warning Signals.

6 Toy Models and the Search for a Physical Mechanism

The failure of the accretion-disc stability hypothesis suggested that the appearance of Early Warning Signals cannot be explained solely by the existence of a stable or unstable disc. Instead, it became necessary to identify the specific dynamical ingredient responsible for producing Critical Slowing Down.

The observational results indicated that T CrB experiences a gradual evolution toward eruption, whereas RS Oph appears to undergo a more abrupt transition. This observation motivated the development of a series of simplified toy models designed to isolate the conditions required for EWS to emerge.

The purpose of these models was not to reproduce every aspect of recurrent nova physics, but rather to determine which physical mechanisms are capable of generating the statistical signatures observed in the light curves.

7 Disc Instability Alone Does Not Produce EWS

The first toy models were inspired by thermal-viscous disc instability scenarios.

Disc instabilities can produce dramatic luminosity variations and are often invoked to explain activity in accreting compact objects. It therefore seemed reasonable to investigate whether the onset of instability itself could generate Early Warning Signals.

In these experiments the system was forced to transition rapidly from a stable state into an unstable regime. Such transitions successfully produced outburst-like behavior in the synthetic light curves. However, the EWS analysis yielded an unexpected result.

Although the luminosity changed dramatically, the classical indicators of Critical Slowing Down generally failed to increase consistently. Variance, autocorrelation, DFA, DET, and LAM either remained unchanged or displayed weak and inconsistent trends.

This result demonstrated an important point:

A sudden instability is not sufficient to generate Early Warning Signals.

The system may become unstable, but if the transition occurs too rapidly, there is insufficient time for Critical Slowing Down to develop.

Consequently, the presence of an instability alone cannot explain the EWS observed in T CrB.

8 A Slowly Driven Accretion System

The next step was to investigate whether the rate at which the system approaches instability is more important than the instability itself.

To test this idea, the accretion rate was treated as a slowly evolving control parameter.

The accretion rate was prescribed as

$$\dot{M}(t) = \dot{M}_{\text{stable}} - \epsilon(t - t_0), \quad (12)$$

where

- $\dot{M}_{\text{stable}}$ is the stable accretion rate,
- ϵ controls the rate of evolution,
- t_0 marks the onset of the drift.

This model assumes that the accretion rate decreases gradually over time, driving the system from a stable burning regime toward an unstable one.

A key feature of the model is the separation of timescales.

The control parameter evolves slowly compared with the dynamical response of the system,

$$\left| \frac{d\dot{M}}{dt} \right| \ll \left| \frac{dx}{dt} \right|, \quad (13)$$

allowing the system to remain close to quasi-equilibrium throughout most of its evolution.

This condition is precisely the situation in which Critical Slowing Down is expected to occur.

9 Inclusion of Stochastic Fluctuations

Real accretion flows are inherently noisy.

To account for this effect, a stochastic component was added to the accretion rate evolution:

$$\dot{M}(t) = \dot{M}_{\text{stable}} - \epsilon(t - t_0) + \eta(t), \quad (14)$$

where $\eta(t)$ represents Gaussian white noise.

The stochastic forcing serves two purposes.

First, it mimics the random variability naturally present in accreting systems.

Second, it provides the perturbations required to measure Critical Slowing Down. Without fluctuations, quantities such as variance and autocorrelation cannot be meaningfully evaluated.

The inclusion of noise produced synthetic light curves qualitatively similar to observational data while preserving the slow evolution of the underlying control parameter.

10 Emergence of Early Warning Signals

The slowly driven models produced a strikingly different result from the disc-instability experiments.

As the system approached instability, multiple EWS indicators began to increase simultaneously.

The most consistent trends were observed in

- Variance,
- DFA exponent,
- Determinism (DET),
- Laminarity (LAM).

In some parameter combinations, lag-1 autocorrelation also increased.

The appearance of these trends before the onset of instability strongly suggests that the system experiences Critical Slowing Down during the approach to the transition.

Importantly, the EWS appeared before the outburst itself.

This distinction is crucial.

The indicators are not responding to the eruption. Instead, they are responding to the gradual loss of stability that occurs before the eruption.

The toy models therefore support the interpretation that Early Warning Signals arise from the approach to instability rather than from the instability itself.

11 Timescale Study

Having established that slowly varying accretion rates can generate EWS, the next objective was to determine how the characteristic timescale of the evolution influences the results.

To investigate this question, the accretion-rate evolution was modeled using a sigmoid profile. The sigmoid parameter controls the duration over which the transition occurs and therefore provides a convenient way to vary the driving timescale.

Several values of the sigmoid parameter were explored, corresponding to characteristic timescales ranging from approximately two years to eighteen years.

For each timescale, the complete EWS analysis was repeated using multiple combinations of

- detrending parameters,
- rolling-window sizes,
- recurrence-analysis settings.

The results revealed a clear trend.

For short timescales ($\lesssim 10$ years), most simulations exhibited either weak EWS or no EWS at all. Individual indicators occasionally showed increasing trends, but these trends were generally not robust across different analysis parameters.

Intermediate timescales ($\sim 10 - 12$ years) produced somewhat stronger signatures, although the indicators still lacked consistency.

A dramatic change occurred for timescales near fifteen years.

At approximately

$$\tau \approx 15.1 \text{ yr}, \tag{15}$$

multiple EWS indicators began to increase simultaneously for a substantial fraction of the parameter combinations.

The strongest signatures were observed for

$$\tau \approx 18.1 \text{ yr.} \quad (16)$$

For these models, variance, DFA, autocorrelation, DET, and LAM frequently increased together, providing a coherent EWS signal consistent with Critical Slowing Down.

The robustness of the results across different detrending parameters and window sizes indicates that the observed trends are not artifacts of the analysis procedure.

Instead, they appear to be a genuine consequence of the long-timescale evolution of the control parameter.

12 Interpretation of the Timescale Dependence

The timescale study provides important insight into the physical origin of Early Warning Signals.

Critical Slowing Down requires the system to remain close to equilibrium while stability is gradually lost. If the control parameter changes too rapidly, the system is effectively driven through the transition before the slowing-down process can become observable.

Long driving timescales provide sufficient time for perturbations to accumulate and for the characteristic statistical signatures to emerge.

Consequently, the strength of the EWS increases with the duration of the approach to instability.

This result suggests that the existence of EWS is controlled not by the occurrence of an outburst itself, but by the rate at which the system evolves toward the unstable state.

The observational differences between T CrB and RS Oph may therefore reflect differences in their long-term accretion-rate evolution rather than differences in the nature of their instabilities.

The timescale study thus identifies slow mass-transfer evolution as the most promising candidate mechanism for explaining the appearance of EWS in recurrent novae.

13 Summary of Timescale Dependence

To quantify the influence of the driving timescale on the emergence of Early Warning Signals, we performed a systematic parameter study using sigmoid-driven accretion-rate evolution. Multiple combinations of detrending parameters and rolling-window sizes were explored for each characteristic timescale.

Rather than focusing on individual parameter combinations, we summarize the overall behavior by considering whether multiple EWS indicators simultaneously exhibited increasing trends.

Table 1: Summary of EWS behavior as a function of the characteristic driving timescale.

Timescale (yr)	EWS Strength	Interpretation
1.94	Very Weak	No consistent EWS
5.23	Weak	Occasional increases in variance and RQA metrics
8.52	Weak	No robust multi-indicator behavior
11.80	Moderate	Some indicators begin to increase
15.09	Strong	Multiple indicators increase simultaneously
18.13	Very Strong	Robust EWS across many parameter sets

The results reveal a clear transition near timescales of approximately 15–18 years. Below this threshold, EWS indicators appear only sporadically and are sensitive to analysis parameters. Above this threshold, several independent indicators simultaneously display increasing trends, indicating the emergence of Critical Slowing Down.

The complete parameter-study table is provided in Appendix A.

14 Physical Interpretation

The toy-model results suggest that the appearance of Early Warning Signals is controlled primarily by the rate at which the system approaches instability.

The observational behavior of T CrB provides an intriguing connection to this result.

Recent studies have reported that T CrB entered an extended high-accretion state prior to its anticipated eruption. Observations indicate a prolonged phase of enhanced mass transfer from the red giant companion to the accretion disc. Such behavior naturally introduces a slowly varying control parameter into the system.

Within the framework developed in this work, the accretion rate acts as the control parameter that governs stability. A gradual evolution of the accretion flow allows the system to remain close to quasi-equilibrium while the restoring forces weaken. Under these conditions, perturbations decay increasingly slowly, producing the observed increase in variance, DFA, DET, and LAM.

The timescale study further strengthens this interpretation. The strongest Early Warning Signals emerged when the characteristic evolution timescale was approximately 15–18 years. This is qualitatively consistent with the long-term mass-transfer evolution inferred for T CrB.

The situation in RS Oph appears fundamentally different. Although RS Oph also possesses an accretion disc, the accretion process is dominated by wind accretion from the red giant companion. Such accretion is expected to be more irregular and stochastic than Roche-lobe overflow.

If the effective control parameter evolves rapidly or fluctuates strongly, the system may pass through unstable regions without remaining sufficiently close to equilibrium for Critical Slowing Down to develop. In this case, robust Early Warning Signals would not be expected.

The absence of strong EWS in RS Oph may therefore reflect the nature of its mass-transfer evolution rather than differences in the eruption mechanism itself.

15 Discussion

The results of this study provide evidence that Early Warning Signals can emerge in accreting white-dwarf systems prior to nova eruptions. However, the existence of EWS appears to depend critically on the evolutionary pathway followed by the system.

Our initial hypothesis attributed the difference between T CrB and RS Oph to the stability of their accretion discs. Subsequent investigation revealed that this explanation was insufficient because both systems are capable of experiencing disc instabilities.

The toy models instead indicate that the relevant factor is the timescale associated with the evolution of the control parameter. Slowly driven systems naturally exhibit Critical Slowing Down, whereas rapidly driven systems do not.

This conclusion is consistent with the general theory of critical transitions. Early Warning Signals are expected to appear only when the system remains near equilibrium while stability is gradually lost. If the transition occurs too rapidly, the characteristic signatures do not have sufficient time to develop.

The results also highlight the importance of combining observational data with simplified theoretical models. Observations alone reveal the presence or absence of EWS, but toy models help identify the physical processes responsible for their emergence.

Although the present models are intentionally simplified, they provide a physically plausible explanation for the contrasting behaviors of T CrB and RS Oph.

Future work should incorporate more realistic descriptions of mass transfer, disc evolution, and nuclear burning. Coupling EWS analysis directly to MESA simulations would provide a valuable next step toward connecting the statistical signatures to the underlying stellar physics.

16 Conclusions

In this work, we investigated the presence of Early Warning Signals in recurrent novae using observational light curves and a sequence of simplified accretion-driven toy models.

The principal findings are summarized below:

1. T CrB exhibits strong and consistent Early Warning Signals prior to eruption, including increasing variance, autocorrelation, DFA, DET, and LAM.
2. RS Oph generally lacks similarly robust EWS signatures.
3. An initial hypothesis based on differences in accretion-disc stability failed to explain the observations.
4. Toy models demonstrated that disc instabilities alone do not necessarily generate Early Warning Signals.
5. Slowly driven accretion systems naturally produce Critical Slowing Down and the associated EWS indicators.
6. A systematic timescale study revealed that strong EWS emerge only when the characteristic evolution timescale exceeds approximately 15 years.
7. The strongest and most robust signals occur for timescales near 18 years.
8. These results suggest that long-term mass-transfer evolution is the most likely origin of the EWS observed in T CrB.

Taken together, the observational analysis and theoretical modeling support a unified interpretation in which Early Warning Signals arise from the gradual loss of stability associated with slowly evolving accretion flows. The existence of EWS in T CrB may therefore provide direct evidence that the system undergoes a prolonged approach to instability before eruption.

A Complete Timescale Study

Table 2 contains the complete set of parameter combinations explored in the sigmoid-driven accretion models. The table lists the trends obtained for DFA, variance, autocorrelation, laminarity, and determinism as functions of detrending parameter, rolling-window size, and driving timescale.

Table 2: Summary of EWS trends for different sigmoid timescales, detrending parameters, and rolling-window sizes. NT = No Trend, Inc = Increasing, Dec = Decreasing.

Sigmoid	Timescale (yr)	Detrend	Window	DFA	Variance	ACF(1)	LAM	DET
100	1.94	0.30	300	NT	NT	NT	NT	NT
100	1.94	0.30	500	NT	NT	NT	NT	NT
100	1.94	0.30	700	NT	NT	NT	Dec	NT
100	1.94	0.30	900	Dec	Inc	NT	NT	NT
100	1.94	0.15	900	Dec	Inc	NT	NT	NT
100	1.94	0.40	300	NT	NT	NT	NT	NT
100	1.94	0.40	500	Dec	NT	NT	NT	NT
100	1.94	0.40	700	NT	NT	NT	Dec	NT
100	1.94	0.40	900	Dec	Inc	NT	NT	Inc
100	1.94	0.50	300	NT	NT	NT	NT	NT
100	1.94	0.50	500	NT	NT	NT	NT	NT
100	1.94	0.50	700	NT	Inc	NT	NT	NT
100	1.94	0.50	900	NT	Inc	NT	NT	Inc
300	5.23	0.30	300	NT	Inc	NT	Inc	Inc
300	5.23	0.30	700	NT	Inc	NT	Inc	NT
300	5.23	0.30	900	Dec	Inc	NT	NT	NT
300	5.23	0.30	1100	NT	NT	NT	NT	NT
300	5.23	0.40	300	NT	Inc	NT	Inc	NT
300	5.23	0.40	700	NT	Inc	NT	Inc	NT
300	5.23	0.40	900	NT	Inc	NT	NT	Dec
300	5.23	0.40	1100	NT	NT	NT	NT	NT
300	5.23	0.50	300	NT	Inc	NT	NT	NT
300	5.23	0.50	700	NT	Inc	NT	Inc	NT
300	5.23	0.50	900	NT	Inc	NT	NT	Dec
300	5.23	0.50	1100	NT	NT	NT	NT	NT
500	8.52	0.30	300	NT	NT	NT	NT	NT
500	8.52	0.30	700	NT	NT	NT	NT	NT
500	8.52	0.30	1000	NT	Dec	NT	NT	NT
500	8.52	0.30	1200	NT	NT	NT	NT	NT
500	8.52	0.40	300	NT	NT	NT	NT	NT
500	8.52	0.40	700	Dec	NT	NT	NT	NT
500	8.52	0.40	1000	NT	NT	NT	NT	NT

Sigmoid	Timescale (yr)	Detrend	Window	DFA	Variance	ACF(1)	LAM	DET
500	8.52	0.40	1200	NT	NT	NT	NT	NT
500	8.52	0.50	300	NT	NT	NT	NT	NT
500	8.52	0.50	700	NT	NT	NT	NT	NT
500	8.52	0.50	1000	NT	NT	NT	NT	NT
500	8.52	0.50	1200	NT	NT	NT	NT	NT
700	11.80	0.30	300	Dec	NT	NT	NT	NT
700	11.80	0.30	700	NT	NT	NT	NT	NT
700	11.80	0.30	1000	Dec	NT	NT	NT	NT
700	11.80	0.30	1200	NT	NT	NT	NT	NT
700	11.80	0.40	700	NT	Inc	NT	NT	NT
700	11.80	0.40	1000	NT	Inc	NT	Dec	Dec
700	11.80	0.40	1200	NT	NT	NT	Inc	NT
700	11.80	0.50	1000	Dec	Inc	Dec	Dec	Dec
700	11.80	0.50	1200	NT	Inc	Dec	Inc	Inc
900	15.09	0.30	300	NT	NT	NT	NT	NT
900	15.09	0.30	700	NT	Inc	NT	NT	NT
900	15.09	0.30	1000	NT	Inc	NT	Inc	NT
900	15.09	0.30	1200	Inc	Inc	Inc	Inc	Inc
900	15.09	0.40	300	NT	NT	NT	NT	NT
900	15.09	0.40	700	NT	Inc	NT	NT	NT
900	15.09	0.40	1000	NT	Inc	NT	NT	NT
900	15.09	0.40	1200	Inc	Inc	Inc	Inc	Inc
900	15.09	0.50	300	NT	NT	NT	NT	NT
900	15.09	0.50	700	NT	NT	NT	NT	NT
900	15.09	0.50	1000	NT	NT	NT	NT	Inc
900	15.09	0.50	1200	NT	Inc	NT	Inc	Inc
1100	18.13	0.30	300	NT	Inc	Inc	Inc	Inc
1100	18.13	0.30	700	NT	Inc	NT	NT	NT
1100	18.13	0.30	1000	Inc	Inc	Inc	NT	NT
1100	18.13	0.30	1200	Inc	Inc	Inc	NT	NT
1100	18.13	0.30	1400	Inc	NT	NT	NT	NT
1100	18.13	0.40	300	NT	Inc	Inc	Inc	Inc
1100	18.13	0.40	700	NT	NT	NT	NT	NT
1100	18.13	0.40	1000	Inc	Inc	Inc	Inc	Inc
1100	18.13	0.40	1200	Inc	Inc	Inc	Inc	Inc
1100	18.13	0.40	1400	Inc	Dec	Inc	Dec	Dec

Sigmoid	Timescale (yr)	Detrend	Window	DFA	Variance	ACF(1)	LAM	DET
1100	18.13	0.50	300	NT	NT	NT	NT	NT
1100	18.13	0.50	700	Dec	Dec	Dec	Dec	Dec
1100	18.13	0.50	1000	Inc	Inc	NT	Inc	Inc
1100	18.13	0.50	1200	Inc	NT	Inc	Inc	Inc
1100	18.13	0.50	1400	Inc	Inc	Dec	Dec	Dec

References

- Richard D. Alexander, Graham A. Wynn, Andrew R. King, and James E. Pringle. Disc instability in rs ophiuchi: a path to type ia supernovae? *Monthly Notices of the Royal Astronomical Society*, 418(4):2576–2587, 2011. doi: 10.1111/j.1365-2966.2011.19673.x.
- D. A. Bollimpalli, J. M. Hameury, and J. P. Lasota. Disc instabilities and nova eruptions in symbiotic systems: Rs ophiuchi and z andromedae. *Monthly Notices of the Royal Astronomical Society*, 481(4):5422–5438, 2018. doi: 10.1093/mnras/sty2636.
- S. V. George, Sneha Kachhara, Ranjeev Misra, and G. Ambika. Early warning signals indicate a critical transition in betelgeuse. *Astronomy & Astrophysics*, 640:L21, 2020. doi: 10.1051/0004-6361/202038462.
- Marten Scheffer, Jordi Bascompte, William A. Brock, Victor Brovkin, Stephen R. Carpenter, Vasilis Dakos, Hermann Held, Egbert H. van Nes, Max Rietkerk, and George Sugihara. Early-warning signals for critical transitions. *Nature*, 461(7260):53–59, 2009. doi: 10.1038/nature08227.